

Activity

Group members:

Grap applications data

We are now interested in testing whether there is a relationship between the *rank* of a student's undergraduate institution (1 = most selective, 4 = least selective) and their acceptance to a graduate program, after accounting for GRE score and undergraduate GPA.

We fit the regression model

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

Questions

1. Write down the null and alternative hypotheses corresponding to the research question.
2. Write down the reduced model.

3. Below is output for the fitted models. Calculate an appropriate test statistic to test the hypotheses in question 1. What will be the degrees of freedom for the χ^2 distribution under H_0 ?

```
Call: glm(formula = admit ~ gpa + gre + factor(rank), family = binomial,  
data = grad_app)
```

Coefficients:

(Intercept)	gpa	gre	factor(rank)2	factor(rank)3
-3.989979	0.804038	0.002264	-0.675443	-1.340204
factor(rank)4				
-1.551464				

Degrees of Freedom: 399 Total (i.e. Null); 394 Residual

Null Deviance: 500

Residual Deviance: 458.5 AIC: 470.5

```
Call: glm(formula = admit ~ gpa + gre, family = binomial, data = grad_app)
```

Coefficients:

(Intercept)	gpa	gre
-4.949378	0.754687	0.002691

Degrees of Freedom: 399 Total (i.e. Null); 397 Residual

Null Deviance: 500

Residual Deviance: 480.3 AIC: 486.3